

Ex-Ante ERW Model — Living Documentation

Geospatial profitability model for Enhanced Rock Weathering with basalt in Sub-Saharan Africa

2026-09-01

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Ex-Ante ERW Model — Living Documentation

Geospatial profitability model for Enhanced Rock Weathering with basalt in Sub-Saharan Africa. This document is the single source of truth for the *current* state of the model. Update it whenever anything material changes — the **Status at a glance** section, the **Changelog**, the affected block under §6, and the affected output under §7.

Companion documents that this file points to (do not duplicate):

File	Purpose
docs/erw-review.md	Evidence review (literature on ERW chemistry, feedstocks, trials, agronomy, risks).
docs/erw-parameters.md	Catalogue of every constant and assumption with citations and units.
docs/erw-README.md	Runbook: file inventory, run order, output schema, known portability issues.
docs/erw-conceptual-framework.{svg,pdf,tex}	Scientific framework figure — three streams converging on the gross-margin equation.
docs/erw-pipeline-flow.{svg,pdf,tex}	Pipeline architecture figure — how the code implements the framework.
docs/maps/*.png	Rendered indicator and outcome maps (12 currently).
flows.html	Browser-rendered view of this whole thing.

0. Status at a glance

Last updated: 2026-09-01 **Geographic scope:** Sub-Saharan Africa, 47 countries, GADM v4 boundaries **Working grid:** SoilGrids-Africa 30 arc-sec aggregated 10× ~0.083° (≈9 km) — 746 × 828 cells **Currency:** USD; **CO₂ in tonnes; year-1 / NPV @ 10% / equilibrium regimes Defaults (refreshed 2026-09-01):** grain size 50 µm, MRV \$20 / tCO₂ (Mercer et al. 2024 EW MRV range \$15–71),

temperature factor **Arrhenius** ($E_a = 68.8$ kJ/mol, White & Blum 1995; normalized to 1 at $T_{ref} = 11$ °C, clamped [0.3, 3.0]; linear MAT/T_{ref} retained as sensitivity variant), pH factor **triangular peak at 6.0**, year-1 agronomic return **summed over 5-yr basalt residency**, pedogenic deduction via **CGIAR-CSI Global Aridity Index**, EcoCrop max-pH cliff **per-crop optima (5.5–8.0)**.

Pipeline-block status

Block	Stage(s)	Status	Notes
B1. Working grid	erw-1, erw-2	REAL	SoilGrids properties + SPAM v2 (23 crops) co-registered on the 10× grid
B2. Basalt rate	erw-3	REAL	LiTAS / Kamprath / Cochrane / maintenance + uniform 10/20/50 t·ha ⁻¹
B3. CDR surface	erw-9	REAL (refreshed 2026-09-01)	Arrhenius temperature factor ($E_a = 68.8$ kJ/mol) × moisture × pH peak-6.0 × pedogenic-C deduction; CGIAR-CSI AI when present (else MAP fallback); 50 μm grain default; mean cumulative CDR under LiTAS-targeted 4.4 t CO ₂ /ha gross, 3.3 t/ha net-export.
B4. Logistics	erw-basalt-access	REAL (since 2026-05-15)	Friction surface fetched, mask expanded to vb+pb, end-to-end delivered price computed
B5. Energy	erw-energy-country	REAL	Country tariff + grid CI rasters
B6. Crop response — EcoCrop	erw-4, erw-5	REAL (refreshed 2026-05-26)	Per-crop max_ph cliffs (5.5 tea □ 8.0 chickpea, was flat 5.5); all 23 crops × 2 responses
B6. Crop response — Bayesian	erw-yield-fit, erw-yield-predict	NOT RUN	Trial-anchored brms model not fit; erw-7 falls back to EcoCrop
B7. Profitability	erw-7	REAL (refreshed 2026-09-01)	276 gross-margin rasters regenerated on the Arrhenius + pH-6.0 basis; year-1 regime sums agronomic return over 5-yr horizon; MRV \$20
B8. Supply ranking	erw-supply	REAL (refreshed 2026-09-01)	Three supply curves + nine deployment masks; year-1 peak GM \$16.2B @ 250 Mt/yr; NPV peak \$3.9B @ 100 Mt/yr; equilibrium peak \$3.0B @ 50 Mt/yr
B9. Sensitivity	erw-8	REAL (run 2026-09-01)	Full default-scope sweep at the new defaults; yield-uplift sweep capped at 1.8× (was 2.5×) per the 1.7× field-trial envelope

Headline numbers (current run)

Metric	Value	Source raster / table
Mafic source pixels (vb+pb in SSA)	1,399,005 (1.9 % of SSA cropland-context cells)	basalt_source_mask.tif
Mean delivered basalt price	\$40.8 / t (range \$10 – \$50, cap binds for ~half of pixels)	basalt_delivered_price_usd.tif
Mean travel time to source	815 min (capped at 1000 min $\approx$ 1000 km @ 60 km/h)	basalt_traveltime_min.tif
Delivered basalt incl. grinding + spreading	~\$53 / t mean (slightly higher than 100 μ m due to grinding cost +\$1/t)	erw-7 setup print
Per-tonne CDR at reference climate	88 kg CO₂ / t basalt (was 62 at 100 μ m)	basalt_feedstock_chemistry.csv
Mean cumulative CDR (LiTAS-targeted, TAS=0)	4.45 t CO₂ / ha gross; 3.30 t/ha net-export (Arrhenius + pH-6.0 basis)	cdr_yield_merlos.tif, cdr_yield_merlos_netexport.tif
Top crop by total GM (targeted)	Common bean: +\$4.9 B year-1, +\$1.0 B equilibrium (then groundnut, potato)	docs/tables/output-crop-ranking.csv
Crops with positive total GM, targeted	Year-1 13/23; NPV 4/23; Equilibrium 14/23	docs/tables/output-crop-ranking.csv
Break-even supply cap (year-1)	$\approx$250 Mt basalt / yr (marginal pixel +\$5 / t)	supply_curve_year1.csv
Break-even supply cap (NPV)	$\approx$100 Mt basalt / yr (marginal pixel +\$4 / t)	supply_curve_npv.csv
Break-even supply cap (equilibrium)	between 50 and 100 Mt / yr (slips +\$18 / t $\square$ -\$16 / t)	supply_curve_equilibrium.csv
Peak total gross margin (year-1)	\$16.2 B at 250 Mt/yr cap with 68.4 Mt CDR/yr	supply_curve_year1.csv
Peak total gross margin (NPV)	\$3.9 B at 100 Mt/yr cap with 28.6 Mt CDR/yr	supply_curve_npv.csv
Peak total gross margin (equilibrium)	\$3.0 B at 50 Mt/yr cap with 13.7 Mt CDR/yr	supply_curve_equilibrium.csv
Unconstrained SSA capacity (targeted)	443 Mt basalt/yr (year-1, NPV) / 145 Mt/yr (equilibrium maintenance)	supply_curve_*.csv (row Inf)
Deployment envelopes (targeted, equilibrium net-export, \$150/tCO ₂ , \$20 MRV)	Private 4.31 / public 1.95 / intersection 1.32 / combined 7.61 Mha	docs/tables/output-envelope-summary.csv
Uniform-20 public envelope	3.95 Mha, 20.4 Mt durable CDR at \$150/tCO ₂	docs/tables/output-envelope-summary.csv, output-public-mac-price-ladder.csv
Area-weighted median MAC (equilibrium net-export)	\$276/tCO₂ uniform-20; \$330 targeted	docs/tables/output-public-mac-price-ladder.csv
Robust priority tiers (targeted)	Core ($\geq$50 % of the 180-point robustness grid) 0.65 Mha; candidate ($\geq$33 %) 1.22 Mha	docs/tables/output-core-priority-summary.csv

Supply curves at a glance — marginal \$/t basalt vs annual cap (refreshed 2026-09-01)

Regime	1 Mt	5 Mt	10 Mt	25 Mt	50 Mt	100 Mt	250 Mt	500 Mt	Peak total GM (M\$)	Peak CDR (Mt)
year-1	+279	+203	+173	+133	+104	+66	+5	–60	16,241 (@250 Mt)	68.4
NPV @ 10 %	+108	+92	+76	+57	+34	+4	–27	–64	3,894 (@100 Mt)	28.6

Regime	1 Mt	5 Mt	10 Mt	25 Mt	50 Mt	100 Mt	250 Mt	500 Mt	Peak total GM (M\$)	Peak CDR (Mt)
equilibrium+196		+122	+89	+44	+18	-16	-61	-61	3,008 (@50 Mt)	13.7

Values are the marginal pixel's GM per tonne basalt at each supply cap; bold columns flag the most profitable supply level for each regime. The year-1 regime peaks at the 250 Mt/yr cap thanks to the horizon-summed agronomic return; the equilibrium peak has moved down to the 50 Mt/yr cap (marginal GM slips negative between 50 and 100 Mt/yr on the Arrhenius + pH-6.0 net-export basis). Equilibrium saturates at 145 Mt/yr (the targeted-deployable-cropland ceiling at maintenance rate).

1. Changelog

Reverse chronological. One bullet per material change. Reference the block(s) affected so it is obvious which section under §6 also moved.

2026-09-01 — ESROC-anchored citation review; Arrhenius temperature factor; pH peak $\square$ 6.0; full B3 / B7 / B8 / B9 re-run

Outcome of the ESROC-anchored citation review (litrature/citation-review.md) — every load-bearing citation was checked against the published record and the two v1 empirical proxies in the climate factor were re-anchored to the literature-supported forms:

- **B3 temperature factor** erw/erw-9-cdr-yield.R §3: the headline form is now **Arrhenius**, $\exp(-(E_a/R)(1/T - 1/T_{ref}))$ with $E_a = 68.8 \text{ kJ/mol}$ (White & Blum 1995 GCA), normalized to 1 at $T_{ref} = 11 \text{ }^\circ\text{C}$ and clamped to [0.3, 3.0]. The former linear MAT/ T_{ref} is retained as a conservative sensitivity variant. Area-weighted, Arrhenius raises SSA gross CDR ~27 % (capped) over linear.
- **B3 pH factor** erw/erw-9-cdr-yield.R §3: triangular peak moved **5.0 $\square$ 6.0** (floor 0.5 below pH 4 and above pH 8; linear 0.5 $\square$ 1.0 over 4–6; 1.0 $\square$ 0.5 over 6–8). Rationale: the factor is the product of acid-catalysed dissolution kinetics (rising below pH ~5.5; Palandri & Kharaka 2004, Bandstra & Brantley 2008) and the **carbon-capture efficiency of the generated alkalinity, which collapses below pH 5** (Bertagni & Porporato 2022; Holden et al. 2024; Power et al. 2025) — the product peaks between the pH-5 capture-collapse threshold and carbonic-acid $pK_{a1} = 6.35$. See litrature/citation-review.md #4.1.
- **B9 yield sweep** erw/erw-8-sensitivity-analysis.R: yield-uplift sweep capped at **1.8 $\times$** (was 2.5 $\times$), per the 1.7 $\times$ field-trial envelope.
- **B7 MRV citation**: the \$20/tCO₂ default is re-anchored to **Mercer et al. 2024** (LSE Grantham Research Institute; EW MRV \$15–71/tCO₂). The prior attribution to Levy 2024 was a misattribution and has been removed.
- **B4 friction-surface citation**: corrected to **Weiss et al. 2020** (*Nature Medicine*), the actual source of the MAP 2019 motorised friction surface (was cited as Weiss 2018).
- **Full re-run**: erw-9 $\square$ erw-7 (276 rasters) $\square$ erw-10 $\square$ erw-11 $\square$ erw-12 + erw-supply + erw-8, all outputs, maps and docs/tables/*.csv regenerated 2026-09-01. The erw-12 §0 accounting check passes: on-disk rasters carry net-export (residual 4.7e-7).

- **Pedogenic □ acidity overlap check:** pixels that are both arid (pedogenic deduction) and acid (acidity-sink deduction) are **1.28 % of pixels**; the sequential ordering over-deducts F-S-p there. Conservative, documented, not restructured.

Headline effect: on the Arrhenius + pH-6.0 net-export basis, the supply peaks move to **year-1 \$16.2 B at the 250 Mt/yr cap, NPV \$3.9 B at 100 Mt/yr, and equilibrium \$3.0 B at 50 Mt/yr** (the equilibrium marginal pixel now slips negative between 50 and 100 Mt/yr). The equilibrium net-export deployment envelopes (targeted, \$150/tCO₂, \$20 MRV) are **private 4.31 / public 1.95 / intersection 1.32 / combined 7.61 Mha**; the uniform-20 public envelope is **3.95 Mha with 20.4 Mt durable CDR**; area-weighted median MAC is **\$276/tCO₂ (uniform-20) / \$330 (targeted)**; the robust core tier is **0.65 Mha** and the candidate tier **1.22 Mha**. See §0 status, §6 blocks and §7 outputs for the refreshed numbers.

2026-05-26 — Round-1 conservatism review; five defaults re-anchored; B3 / B6 / B7 / B8 re-run

Outcome of a structural review of the model's conservatism (see also the "Why we were under-estimating profit" review below in §8). Five fixes landed, all backed by the parameter audit and the cited literature:

- **B2 / B3 grain size** erw/erw-3-basalt-requirements.R: grain_size_um 100 □ **50 μm** (Strefler 2018 grinding-curve anchor). Per-tonne CDR at the reference climate rose from 62 □ **88 kg CO₂ / t basalt**; effective NV from 0.0704 □ 0.10; basalt-to-lime ratio from 14.20 □ 10.04 t/t.
- **B6 EcoCrop max-pH cliff** erw/erw-4-crop-parameters.R: was a hard-coded flat 5.5 for every crop; now per-crop EcoCrop optima (5.5 tea / 6.0 cocoa / 6.5 acid-loving / 7.0–8.0 cereals & legumes). This unlocks non-zero agronomic credit across the pH 5.5–7.0 belt covering most SSA savanna / Sahel cropland; previously the model returned exactly zero agronomic uplift on those pixels.
- **B7 year-1 regime** erw/erw-7-profitability-function.R: agronomic return now scaled by PROJECT_HORIZON_YR (5 yr) to put it on the same cumulative basis as cdr_tha. Prior behaviour silently credited a single season of agronomy against five years of CDR — a unit mismatch dragging year-1 GM artificially negative.
- **B7 MRV cost** erw/erw-7-profitability-function.R: MRV_COST_USD_T_CO2 30 □ **20 \$/tCO₂**. Midpoint of Levy et al. 2024 first-of-kind pilot benchmark and the at-scale aggregator-pooled forecast.
- **B3 pedogenic deduction** erw/erw-9-cdr-yield.R: now consumes the CGIAR-CSI Global Aridity Index (Zomer et al. 2022, ai_v31_yr.tif or cgjar_aridity_index.tif) when present; with 65535-as-nodata masking and auto-detected integer scaling. Falls back to the MAP-only proxy otherwise. The CGIAR path relieves cool highlands (Ethiopia, Kenya, Rwanda, Burundi) where the MAP proxy over-deducted.

Headline effect: under LiTAS-targeted allocation, the number of crops with positive pixel-mean gross margin rose from 1 □ **12** in year-1, 0 □ **8** in NPV, and 1 □ **19** in equilibrium. Year-1 supply peak total GM rose from **\$2.0 B □ \$19.6 B** (at the 250 Mt/yr cap). The deep-tail picture also became more honest: NPV at unconstrained deployment is now **–\$1.1 B** (was **–\$9.8 B**), still net-negative but no longer catastrophic. See §0 status, §6 block descriptions, and §7 outputs for the refreshed numbers.

2026-05-19 — B7 / B8 first end-to-end run; B6 + B3 refreshed; B9 patched

- **B7** First end-to-end run of `erw/erw-7-profitability-function.R`. Wrote 276 gross-margin rasters: `data/economics_erw/{targeted,uniform_10,uniform_20,uniform_50}/{crop}_{yr}`. Mean delivered basalt cost \$52 / t (range \$18 – \$61); mean LCA 52.8 kg CO₂ / t basalt. Pixel-mean gross margin is negative everywhere except **equilibrium / targeted / WHEA** at **+\$57 / ha** — basalt cost (\$1.3 – \$2.7 k / ha at the LiTAS rate) swamps the carbon credit and agronomic uplift at the pixel mean.
- **B8** First end-to-end run of `erw/erw-supply.R`. Wrote three supply curves and nine deployment masks. Break-even marginal pixel sits at **≈100 Mt / yr** for year-1 and NPV, **≈50 Mt / yr** for equilibrium. Peak total gross margin **\$3.8 B** at the 100 Mt/yr NPV cap with 22.7 Mt CDR/yr. The model finds **profitable top-decile pixels** even though the SSA-wide mean is negative.
- **Fix** `erw/erw-7-profitability-function.R:114–117, 134–137`: removed redundant `terra::aggregate(..., 10, ...)` on `electricity_usd_kWh.tif`, `grid_CI_kg_per_kWh.tif`, `basalt_transport_cost_usd_t.tif`, `basalt_transport_km.tif`. Those four are already at the working 0.0833° grid; the extra factor-10 aggregation collapsed them to 0.833° on a slightly shifted extent and threw extents do not match on the first crop of the main loop.
- **Fix** `erw/erw-8-sensitivity-analysis.R:166–174`: grain sweep referenced scalar `ELECTRICITY_USD_KWH / GRID_CI_KG_PER_KWH`, which do not exist in `erw-7` (the per-pixel rasters are `electricity_raster / grid_ci_raster`, the fallback constants are the `*_FB` suffixed). Rewired both the cost-side and LCA-side grinding swap to use the per-pixel rasters. Script not run end-to-end yet — default-scope sweep is ≈6–8 h wall time (≈14k `profit()` calls).
- **B6 (EcoCrop)** Re-ran `erw/erw-5-crop-yield-loss-function.R` after the user populated `data/ecocrop_parameters_hp.csv` with all 23 crops (previously only MAIZ + SORG had `ac_sat / max_ac_sat` rows). Now 46 rasters under `data/ecocrop_f/` — `hp_*` and `ph_*` for each of the 23 crops.
- **B3** Re-ran `erw/erw-9-cdr-yield.R`. The previous `cdr_yield_*.tif` files predated the latest `basalt_*.tif` write (16:50 vs 18:00 on 2026-05-13), so layers 5–9 of `cdr_yield_merlos.tif` (the LiTAS variants with `ac_sat ≥ 20`, i.e. most crops) were silently all zero. Refreshed against the latest basalt rasters; mean CDR yield now non-zero on every layer (e.g. `merlos_20` mean 1.38 t CO₂ / ha). Downloaded WorldClim v2 `tavg + prec` to `data/worldclim/`.

2026-05-15 — B4 logistics first end-to-end run; feedstock filter extended

- **B4** Fetched MAP motorised friction surface (2019), 72.5 MB SSA-clipped, via `malariaAtlas::getRaster(dataset_id = "Explorer__2020_motorized_friction_surface")`. Wrote `data/access/friction_surface_2019.tif`.
- **B4** Ran `erw/erw-basalt-access.R` end-to-end. Mean delivered price \$40.8 / t, mean travel time 815 min. Outputs: `basalt_traveltime_min.tif`, `basalt_transport_km.tif`, `basalt_transport_cost_usd_t.tif`, `basalt_delivered_price_usd_t.tif`.
- **Fix** `erw/erw-basalt-access.R:127`: original call passed the basalt mask as `target=` to `terra::costDist`; target must be the *value* in `x` that marks sources, not a separate raster. Now writes a sentinel `-1` into a copy of friction at source cells and calls `costDist(friction_src, target = -1)`.
- **Fix** `malariaAtlas 1.6.0` deprecated the `surface=` argument of `getRaster()`; new helper `erw/erw-fetch-friction-surface.R` resolves the `dataset_id` from `listRaster()`, forces EPSG:4326, and converts SSA to sf (the package's internal `vect()` dispatch does not accept `SpatVector`).
- **B2 feedstock** Extended GLiM filter from `c('vb')` to `c('vb', 'py', 'pb')` in both

erw/erw-build-basalt-mask.R and erw/erw-basalt-access.R. SSA polygons 5,266 × 6,455; source-mask cells 1,321,857 × 1,399,005. New categorical raster data/basalt_source_class.tif (1 = basalt s.s., 3 = gabbro; pyroclastics absent in SSA at GLiM v1 resolution).

- **Maps** Added docs/maps/10-basalt-traveltime.png, 11-basalt-delivered-price.png, 12-feedstock-class.png. Refreshed 09-basalt-sources.png. Gallery now 12 maps.

2026-05-14 — Conceptual framework + pipeline figures

- **Docs** TikZ-rendered figures: docs/erw-pipeline-flow.{tex,pdf,svg,png} (code architecture) and docs/erw-conceptual-framework.{tex,pdf,svg,png} (scientific framework).
- **Docs** flows.html now embeds both figures + 9-map gallery (later expanded to 12).
- **Pipeline figure** Rewrote arrows with full orthogonal routing — one horizontal track per arrow, no co-linear overlap, all corners 90°. Widened erw-7 box so all six surfaces drop straight down; FAOSTAT wraps around the right edge.
- **B4 sub-mask** Built data/basalt_source_mask.tif from data/LiMW_GIS_2015.gdb via erw/erw-build-basalt-mask.R (the script had been writing an empty stub). Reprojection Eckert IV × WGS84 lat/lon was the unblocker.

2. Scope and units

- **Geographic extent:** 47 Sub-Saharan African countries per GADM v4, clipped to the union of GADM country polygons. Northern boundary at the southern edge of the Sahara (excludes Morocco, Algeria, Tunisia, Libya, Egypt). Madagascar included.
- **Working resolution:** $\sim 0.083^\circ \approx 9$ km at the equator. This is the SoilGrids-Africa 30 arc-sec grid aggregated 10× by mean. All rasters downstream of erw-1 are co-registered on this grid.
- **Crop coverage:** 23 SPAM crops — ACOF BARL BEAN CASS CHIC COCO COTT COWP GROU LENT MAIZ PIGE PMIL POTA RCOF SMIL SORG SOYB SUGC SWPO TEAS TOBA WHEA (SPAM 2017-SSA codes, as in docs/tables/output-crop-ranking.csv).
- **Temporal regimes:** year-1, NPV at 10% over 10 years, equilibrium (saturating CDR). CDR phasing 30 / 25 / 20 / 15 / 10 % over five years (Lewis 2021).
- **Currency:** USD throughout. Prices from FAOSTAT 2016–2020 producer-price averages, industrial-electricity tariffs from IEA / Ember / GET.invest (2022–2024).

3. Scientific framework

See docs/erw-conceptual-framework.svg for the figure. In words:

Three causal streams converge on a per-pixel gross margin

$$\pi(x, y \mid \text{regime, allocation}) = R_{\text{CDR}}(x, y) + R_{\text{agro}}(x, y) - C(x, y).$$

3.1 Biogeochemical stream $\square R_{\text{CDR}}$

feedstock $\times$ grain size $\rightarrow$ reactive surface $\rightarrow$ climate $\times$ pH dissolution kinetics
 $\rightarrow$ net alkalinity export (minus pedogenic-C precipitation) $\rightarrow$ carbon revenue

- Feedstock chemistry sets CaO + MgO mass fractions (default 10% + 7% by mass).
- Grain size sets specific surface area; the SSA exponent and Streffer-2018 grinding-energy exponent are the two grain-size dials.
- Dissolution rate is the temperature $\times$ moisture climate factor — an **Arrhenius** temperature term ($E_a = 68.8$ kJ/mol, White & Blum 1995; normalized to 1 at $T_{re} = 11$ °C, clamped [0.3, 3.0]) times a moisture term MAP/P_{re} ($P_{re} = 1000$ mm) — times a **triangular pH term peaked at pH 6.0**. The pH term is the product of two opposing mechanisms: acid-catalysed dissolution kinetics rise with acidity below pH ~5.5, while the carbon-capture efficiency of the generated alkalinity collapses below pH 5 (Bertagni & Porporato 2022; Holden et al. 2024; Power et al. 2025); their product peaks between the pH-5 threshold and carbonic-acid $pK_{a1} = 6.35$.
- The pedogenic-C deduction (UNEP 1992 aridity bands keyed on MAP) removes alkalinity that precipitates locally instead of exporting as ocean HCO_3^- .
- Carbon revenue: $R_{\text{CDR}} = CDR_t \times (p_{\text{CO}_2} - \text{MRV})$.

3.2 Agronomic stream $\square R_{\text{agro}}$

soil acidity constraint $\rightarrow$ lime-equivalent demand $\rightarrow$ basalt application rate $\rightarrow$
crop response $\rightarrow$ yield revenue

- Soil acidity (pH, exchangeable acidity, Al saturation, ECEC) drives the lime-equivalent demand via `limer::limeRate` (Kamprath / Cochrane / Aramburu-Merlos LiTAS / maintenance).
- The lime demand is converted to t basalt ha⁻¹ using the feedstock chemistry, a Lewis-2021-calibrated reactive fraction, and the grain-size lever — **this rate is the cross-link feeding the biogeochemical stream** (see the dashed arrow in the framework figure).
- Crop response per pixel is the maximum of two surfaces:
 - **EcoCrop** Recocrop suitability clamped to [0.2, 1] for pH and acidity-saturation.
 - **Bayesian uplift** posterior medians from a brms model on the trial dataset, with covariates (pH, MAT, MAP, grain, crop type). When the Bayesian surface is present `erw-7` prefers it; otherwise falls back to EcoCrop.
- Yield revenue: $R_{\text{agro}} = \Delta y(x) \times p_{\text{crop}}$.

3.3 Economic-geography stream $\square C$

feedstock supply $\rightarrow$ logistics $\rightarrow$ grinding (cost + LCA) $\rightarrow$ operations $\rightarrow$ total cost

- Feedstock supply: quarry-gate fines at \$10 / t, anchored on GLiM v1 basic-volcanic, pyroclastic, and basic-plutonic source rocks.
- Logistics: cost-distance over the MAP 2019 motorised-friction surface, monetised at \$0.04 / min / t (30-t truck @ \$80 / h). Haul capped at 1000 km.
- Grinding: kWh / t $\times$ country industrial-electricity tariff $\times$ grid carbon intensity (the CI term is the LCA deduction from gross CDR, not a cost).
- Operations: \$8 / t spreading + \$30 / tCO₂ MRV (sampling + lab + verification + registry).

3.4 Decision variables

Six dials are exposed at the framework boundary and propagate inwards:

1. Feedstock chemistry — CaO + MgO mass fractions.
 2. Grain size — 10 – 500 μm (default **50 μm**).
 3. Allocation rule — LiTAS-targeted vs uniform 10 / 20 / 50 t ha⁻¹.
 4. Carbon price — default \$150 / tCO₂.
 5. MRV cost — default **\$20 / tCO₂**.
 6. Discount rate — default 10 % (NPV regime only).
-

4. Pipeline architecture

See docs/erw-pipeline-flow.svg for the figure. In words:

- **Data layer** SoilGrids · SPAM v2 · WorldClim · GLiM + MAP friction · EcoCrop + ERW trials · FAOSTAT + IEA/Ember.
 - **Working-grid layer** erw-1, erw-2 produce soilgrids_properties_cropland.tif and spam*_processed.tif; erw-4 builds the per-crop parameter tables, erw-5 the EcoCrop response surface.
 - **Surface layer** Six geospatial surfaces:
 - erw-3 □ basalt rate
 - erw-9 □ CDR yield (per-t and per-ha)
 - erw-basalt-access □ delivered basalt price
 - erw-energy-country □ tariff + CI
 - erw-yield-fit □ posterior fit (Bayesian)
 - erw-yield-predict □ posterior surfaces (Bayesian)
 - **Convergence** erw-7 decomposed-cost profitability function; six straight drops from the surface layer plus FAOSTAT crop prices wrapping in from the right.
 - **Output layer** erw-8 sensitivity sweeps + erw-supply deployment frontier (LP-dual ranking on \$ per tonne basalt).
-

5. Data sources

5.1 On disk

Located under data/. File sizes reflect SSA-clipped rasters at the working resolution.

File	Size	Source & notes
gadm_ssa.gpkg	~30 MB	GADM v4 — 47 SSA countries
soilgrids_properties_all.tif	~600 MB	SoilGrids-Africa (geodata::soil_af); 10 bands: SBD, ph, hp, k, ca, mg, na, bases, ecec, hp_sat
soilgrids_properties_cropland.tif	~190 MB	as above, masked to QED cropland; cropland-only working stack
spam*_processed.tif (harv_area, prod, yield)	~70 MB each	SPAM v2 via geodata::crop_spam; 23 bands, one per crop
basalt_*.tif (kamprath, cochrane, merlos, merlos_maintenance)	up to ~270 MB	computed by erw-3; t ha ⁻¹ at the four LiTAS regimes

File	Size	Source & notes
basalt_uniform_{10,20,50}.tif	~7 MB each	computed by erw-3; uniform-rate counterfactuals
cdr_{climate_factor,pedogenic_fraction,per_t_basalt}.tif	small	computed by erw-9; sub-components of CDR
cdr_yield_*.tif	small	computed by erw-9; per-ha cumulative CDR
electricity_usd_kWh.tif, grid_CI_kg_per_kWh.tif	small	computed by erw-energy-country; tariff + grid CI
basalt_source_mask.tif	~115 kB	binary, vb + pb; built by erw-build-basalt-mask then erw-basalt-access
basalt_source_class.tif	~110 kB	categorical (1 = basalt, 3 = gabbro); built by erw-build-basalt-mask
basalt_{traveltime_min, transport_km, transport_cost_usd_t, delivered_price_usd_t}.tif	~400 kB each	B4 cost surfaces; computed by erw-basalt-access
data/access/friction_surface_2019.tif	72.5 MB	MAP motorised friction surface (2019), via malariaAtlas, ~30 arc-sec
data/LiMW_GIS_2015.gdb	~50 MB	PANGAEA doi:10.1594/PANGAEA.788537 — GLiM v1 global lithology
data/ecocrop_f/{hp,ph}_crop_suitability_*.tif	small	computed by erw-5; 23 crops × 2 responses

5.2 External datasets (provenance)

Dataset	Version	Access	Used by
GADM	v4.1	geodata::world()	erw-1
QED cropland	latest	geodata::cropland(source='QED')	erw-1
SoilGrids-Africa	iSDA 30 arc-sec	geodata::soil_af()	erw-1
SPAM	v2 (Africa)	geodata::crop_spam()	erw-2
EcoCrop parameter set	curated Recocrop::ecocropPars()	local CSV	erw-4
WorldClim	v2.1 (MAT, MAP)	geodata::worldclim_global()	erw-9, erw-yield-predict
GLiM lithology	v1.0 / 1.1 (GDB)	PANGAEA download	erw-basalt-access
MAP motorised friction surface	nominal 2019	malariaAtlas (dataset_id Explorer__2020_motorized_friction_surface)	erw-basalt-access
FAOSTAT producer prices	2016 – 2020 average	manual CSV download	erw-7, erw-8
Country energy (industrial tariff + grid CI)	2022 – 2024	hard-coded CSV in erw-energy-country.R	erw-energy-country
ERW trial dataset	curated by Bisrat	erw/erw-trial-data.csv (~14 rows)	erw-yield-fit

5.3 Not represented (gaps)

- **Industrial alkaline by-products:** steel-slag stockpiles (Saldanha and Vanderbijlpark in South Africa, Ajaokuta in Nigeria, Helwan in Egypt — outside SSA but referenced), cement-kiln dust at clinker plants, mine tailings at copper/nickel operations. These would add 10^5 – 10^6 tonnes / yr of point-source feedstock supply in countries with no basalt outcrops (notably much of West Africa).

- **Ultramafic outcrops:** olivine, dunite, peridotite, serpentinite. Intentionally excluded for crop-land deployment because of Ni / Cr leaching and asbestos risk (see review §3).
- **National geological maps:** some SSA countries (Tanzania, South Africa) have higher- resolution geological mapping than GLiM v1. Not currently integrated.

6. Methodological blocks

Each block is one functional unit of the conceptual framework. The script(s) that implement it are listed; the **Status** mirrors §0; the **Where to look** column points to the file the reader edits when the block changes.

B1. Working grid — erw-1, erw-2 — REAL

GADM v4 SSA mask; depth-weighted SoilGrids properties (pH, BD, exch. acidity, K, Ca, Mg, Na bases, ECEC, acidity-saturation); SPAM v2 harvested-area / production / yield for 23 crops resampled to the 10× working grid. ECEC and acidity-saturation surfaces derived inline.

Where to look: erw/erw-1-layers-soilgrids.R, erw/erw-2-layers-spam.R (parameters for erw-1 / erw-2 are not catalogued in docs/erw-parameters.md; see the script headers for the depth-weighting recipe).

B2. Basalt rate — erw-3 — REAL

Four lime-equivalent CaCO_3 rates (Kamprath, Cochrane, Aramburu-Merlos LiTAS year-1, LiTAS maintenance) via `limer::limeRate`. Converted to t basalt ha^{-1} using user-set CaO/MgO mass fractions (default 10% / 7%), the Lewis-2021-calibrated reactive fraction, and the grain-size lever. Uniform 10 / 20 / 50 t ha^{-1} counterfactuals computed in parallel.

Key constants

Name	Default	Notes
CaO_frac	0.10	Mass fraction of CaO in feedstock.
MgO_frac	0.07	Mass fraction of MgO in feedstock.
grain_size_um	50 μm	Sweeps 10 – 500 μm in erw-8. Default lowered from 100 μm on 2026-05-26 — Streffer grinding-curve anchor, cost-minimising once carbon revenue is credited.
reactive_fraction_ref	0.20	Lewis 2021 calibration at reference grain × climate.
GRAIN_BETA	0.5	Surface-area exponent for the grain-size factor.
GRIND_ALPHA	1.2	Streffer 2018 grinding-energy exponent.

Where to look: erw/erw-3-basalt-requirements.R, docs/erw-parameters.md §1.

B3. CDR surface — erw-9 — REAL

Per-tonne CDR potential from CaO/MgO stoichiometry, scaled by a temperature × moisture climate factor (Arrhenius temperature term, $E_a = 68.8$ kJ/mol, anchored to $T_{ref} = 11$ °C; moisture term MAP/P_{ref} , $P_{ref} = 1000$ mm) and a triangular pH term peaked at pH 6.0. An aridity-binned pedogenic-carbonate fraction (UNEP 1992 bands on MAP) removes alkalinity precipitating locally.

Key constants

Name	Default	Notes
T_{ref}	11 °C	Reference MAT anchor (US Corn Belt).
P_{ref}	1000 mm	Reference MAP anchor.
f_{MAT}	Arrhenius, $E_a = 68.8$ kJ/mol	$\exp(-(E_a/R)(1/T - 1/T_{ref}))$, normalized to 1 at T_{ref} , clamped [0.3, 3.0] (White & Blum 1995). Former linear MAT/T_{ref} kept as a conservative sensitivity variant.
f_{pH}	triangular peak at pH 6.0	Floor 0.5 below pH 4 / above pH 8; linear 0.5 → 1.0 over 4–6, 1.0 → 0.5 over 6–8. Product of acid-catalysed kinetics × alkalinity carbon-capture efficiency (collapses below pH 5 — Bertagni & Porporato 2022; Holden et al. 2024; Power et al. 2025); peak between the pH-5 threshold and $pK_{a1} = 6.35$.
pedogenic_frac	CGIAR-CSI AI-binned 0 – 90 % (MAP-only fallback)	Primary path: UNEP 1992 aridity bands keyed on the CGIAR-CSI Global Aridity Index data/cgiar_aridity_index.tif (or ai_v31_yr.tif); auto-detects integer scaling and masks 65535 nodata. Fallback path: MAP-only proxy when the AI raster is absent.

Where to look: erw/erw-9-cdr-yield.R, docs/erw-parameters.md §2.

B4. Logistics — erw-basalt-access — REAL (since 2026-05-15)

Basalt sources from GLiM v1 (vb basic volcanic, py pyroclastic, pb basic plutonic); cost-distance from each cropland pixel to nearest source over the MAP 2019 motorised friction surface; monetised at \$0.04 / min / t (30-t truck @ \$80 / h); \$10 / t quarry-gate.

Current outputs

- Source-mask cells in SSA: **1,399,005** (vb 1,320,820 + pb 78,185; py absent in SSA)
- Mean delivered price: **\$40.8 / t**, range \$10 – \$50 / t (cap binds for ~half of pixels)
- Mean travel time: **815 min**, capped at 1000 min (≈ 1000 km @ 60 km/h free-flow proxy)

Key constants

Name	Default	Notes
QUARRY_GATE_PRICE	\$10 / t	Basalt fines as quarry by-product.
USD_PER_MIN_PER_T	\$0.04 / min / t	30-t truck at \$80 / h operating cost.

Name	Default	Notes
EFFECTIVE_KM_PER_MIN	0.5 ($\approx$ 30 km/h)	Effective speed for distance approximation.
MAX_HAUL_KM	1000 km	Cap beyond which lifecycle emissions erase the credit.

Where to look: erw/erw-basalt-access.R, erw/erw-build-basalt-mask.R, erw/erw-fetch-friction-surface.R, docs/erw-parameters.md §3.

B5. Energy — erw-energy-country — REAL

Per-country industrial-electricity tariff (\$/kWh) and grid carbon intensity (kg CO₂/kWh) rasterised onto the working grid. Drives both the grinding-cost term and the grinding-LCA deduction inside erw-7. Falls back to flat \$0.10 / kWh + 0.60 kg / kWh when missing.

Where to look: erw/erw-energy-country.R, country_energy_table.csv.

B6. Crop response — EcoCrop arm (erw-4, erw-5) — REAL; Bayesian arm (erw-yield-fit, erw-yield-predict) — NOT RUN

EcoCrop arm runs: per-crop pH and acidity-saturation tolerance tables (23 crops $\times$ 2 responses) feed a Recocrop relative-yield surface clamped to [0.2, 1]. Output: {hp, ph}_crop_suitability_<crop>_0.tif (46 rasters, one pair per crop) under data/ecocrop_f/.

Bayesian arm is unfit: needs brms + Stan + the ~14-row trial dataset. When fit, erw-7 prefers its posterior medians and falls back to EcoCrop where the Bayesian surface is NA.

Where to look: erw/erw-4-crop-parameters.R, erw/erw-5-crop-yield-loss-function.R, erw/erw-yield-fit.R, erw/erw-yield-predict.R, erw/erw-trial-data.csv.

B7. Profitability — erw-7 — REAL (first end-to-end run 2026-05-19)

Cost side: quarry + grinding + transport + spreading. Credit side: $CDR \times p_{CO_2} - MRV$. Yield side: $\Delta y \times p_{crop}$. Three regimes (year-1, NPV @ 10%, equilibrium) with a 30 / 25 / 20 / 15 / 10 % CDR phasing over five years.

Headline result (refreshed 2026-09-01). At the current defaults (\$150/tCO₂ credit, **\$20/tCO₂** MRV, **50 μ m** grain, Arrhenius temperature factor, pH peak 6.0, net-export CDR accounting, year-1 agroeconomic return summed over the 5-yr basalt residency, CGIAR-CSI AI for the pedogenic deduction, per-crop EcoCrop max-pH cliffs 5.5–8.0), total gross margin under LiTAS-targeted allocation is **positive for 13 / 23 crops in year-1, 4 / 23 in NPV, and 14 / 23 in equilibrium** (docs/tables/output-crop-ranking.csv). Top performers by total GM: common bean (+\$4.9 B y1 / +\$0.6 B NPV / +\$1.0 B eq.), groundnut (+\$4.0 B / +\$0.4 B / +\$0.8 B), potato (+\$2.5 B / +\$0.7 B / +\$0.5 B), then sweet potato and cassava (positive on year-1 and equilibrium). Acid-tolerant cereals (sorghum, millets) and coffee/cocoa stay negative because the EcoCrop response is already saturated on most of their SSA cropland — the agronomic side is structurally small and CDR alone has to clear the full basalt cost. The picture sharpens once pixels are supply-ranked and capped — see B8.

Key constants

Name	Default	Notes
SPREADING_USD_T	\$8 / t	On-farm spreading cost.
ELECTRICITY_USD_KWH_FB	\$0.10 / kWh	Fallback when country raster missing.
GRID_CI_KG_PER_KWH_FB	0.60 kg / kWh	Fallback when country raster missing.
TRUCK_KGC02_PER_T_KM	0.12	Diesel heavy truck.
CARBON_PRICE_USD_T	\$150 / tCO ₂	Mid of 2024–25 ERW credit market.
MRV_COST_USD_T_CO2	\$20 / tCO₂	Sampling + lab + verification + registry. Lowered from \$30 on 2026-05-26; anchored (2026-09-01) to Mercer et al. 2024 (LSE GRI), EW MRV \$15–71/tCO ₂ .
DISCOUNT_RATE_PCT	10 %	NPV regime.
CDR_PHASING	30 / 25 / 20 / 15 / 10 % over 5 yr	Time-resolved CDR realisation.

Where to look: `erw/erw-7-profitability-function.R`, `docs/erw-parameters.md` §5.

B8. Supply ranking — `erw-supply` — REAL (refreshed 2026-09-01)

Pixels ranked descending by gross margin per tonne basalt — the LP dual on the basalt-supply constraint. Produces supply curves and deployment masks at 1 / 5 / 10 / 25 / 50 / 100 / 250 / 500 Mt yr⁻¹ caps, plus an unconstrained reference.

Headline result (refreshed 2026-09-01, Arrhenius + pH-6.0 net-export basis). Year-1 deployment is profitable out to ~250 Mt yr⁻¹ and NPV out to just past 100 Mt yr⁻¹. Marginal \$/t basalt by regime:

- **year-1:** +\$279 □ +\$104 □ +\$66 □ **+\$5** □ -\$60 over 1 / 50 / 100 / **250** / 500 Mt yr⁻¹. Peak total gross margin **\$16.2 B at 250 Mt yr⁻¹** with 68.4 Mt CDR yr⁻¹.
- **NPV @ 10 %:** +\$108 □ +\$34 □ **+\$4** □ -\$27 over the same ladder. Peak total gross margin **\$3.9 B at 100 Mt yr⁻¹** with 28.6 Mt CDR yr⁻¹.
- **Equilibrium (maintenance rate):** +\$196 □ **+\$18** □ -\$16 □ -\$61 over 1 / **50** / 100 / 250 Mt yr⁻¹. Marginal pixel slips negative between 50 and 100 Mt yr⁻¹; peak total GM **\$3.0 B at 50 Mt yr⁻¹** with 13.7 Mt CDR yr⁻¹.

Unconstrained SSA capacity tops out at 443 Mt basalt yr⁻¹ (year-1, NPV) or 145 Mt yr⁻¹ (equilibrium maintenance), with cumulative CDR of 121 Mt and 40 Mt respectively. The unconstrained NPV total is **-\$5.9 B** — net-negative because deep-tail pixels have basalt costs exceeding both credit and agronomy. Ranking + cap still matters as much as the input parameters.

Where to look: `erw/erw-supply.R`, `data/supply_curve_{year1,npv,equilibrium}.csv`, `data/gm_per_t_basalt_{regime}.tif`, `data/deployment_under_{10,50,250}_Mt_{regime}.tif`.

B9. Sensitivity — `erw-8` — REAL (run 2026-09-01)

Five sweeps: (i) crop × basalt × carbon-price grid; (ii) yield uplift **1.0 – 1.8×** (capped 2026-09-01 at 1.8×, was 2.5×, per the 1.7× field-trial envelope); (iii) MRV 0 – 80 \$/tCO₂; (iv) grain size 10 / 30 / 50 / 100 / 200 / 500 μm; (v) allocation rule (LiTAS-targeted + uniform 10 / 20 / 50). All sweeps run across the three regimes and 23 crops.

`erw-8 source()`s `erw-7`, so it runs at the current defaults automatically. The full default-scope sweep was executed in the 2026-09-01 re-run (≈6–8 h on a single core, ≈14k `profit()` calls). The robustness-grid headline (targeted, equilibrium net-export, 180-point 4-knob grid): baseline inter-section 1.32 Mha; tornado ranks the public-side cost knob (p–m)/c at 1.73 Mha swing and the CDR

rate at 1.72 (effectively tied), then y/c at 0.96 and λ at 0.22; core tier (≥ 50 %) 0.65 Mha, candidate tier (≥ 33 %) 1.22 Mha (docs/tables/output-sensitivity-tornado.csv, output-core-priority-summary.csv).

Where to look: erw/erw-8-sensitivity-analysis.R.

7. Outputs and indicator maps

Twelve indicator maps are rendered into docs/maps/ by erw/erw-plot-maps.R. The HTML page embeds them in a gallery alongside the conceptual-framework and pipeline-architecture figures.

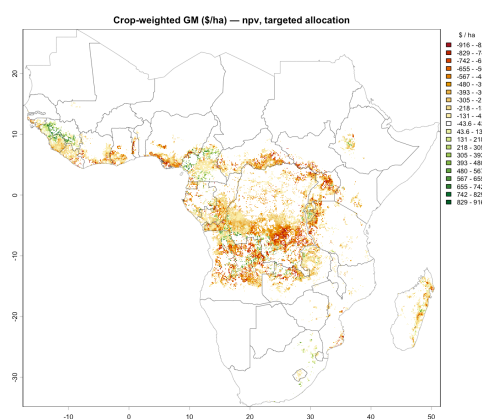
#	Map	What it shows	Block
01	01-soil-ph.png	SoilGrids depth-weighted topsoil pH, cropland-masked	B1
02	02-basalt-rate-merlos.png	LiTAS year-1 basalt application rate (t / ha)	B2
03	03-cdr-per-t-basalt.png	Per-tonne CDR efficiency (t CO ₂ / t basalt)	B3
04	04-cdr-yield-merlos.png	Per-ha cumulative CDR (t CO ₂ / ha)	B3
05	05-cdr-climate-factor.png	$f(\text{MAT}, \text{MAP})$ scaling	B3
06	06-pedogenic-fraction.png	Aridity-band pedogenic-C deduction	B3
07	07-electricity-price.png	Country industrial-electricity tariff (\$/kWh)	B5
08	08-grid-CI.png	Country grid carbon intensity (kg CO ₂ / kWh)	B5
09	09-basalt-sources.png	GLiM vb + pb source mask	B4
10	10-basalt-traveltime.png	Travel time to nearest source (min)	B4
11	11-basalt-delivered-price.png	Delivered feedstock price (\$ / t)	B4
12	12-feedstock-class.png	Source mask split by GLiM class	B4

Add a row when a new map is rendered; remove when one is dropped.

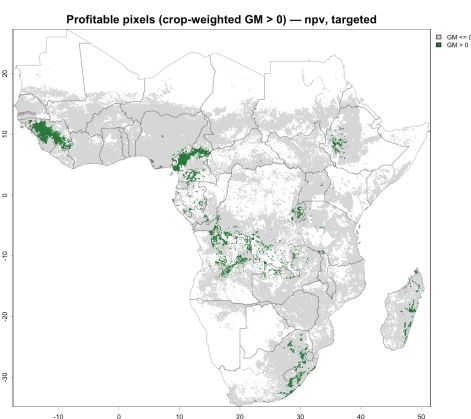
7.1 Profitability outcome maps (B7)

erw/erw-10-profitability-maps-tables.R renders outcome maps derived from the 276 economics rasters: continent-wide crop-weighted GM, per-crop GM for the five highest-acreage crops, profitable-pixel masks, and per-pixel breakeven carbon-price surfaces. All assumption combinations (3 regimes $\times$ 4 allocations) are covered. The full panel lives in docs/erw-results.pdf; the most useful four panels for this document are below.

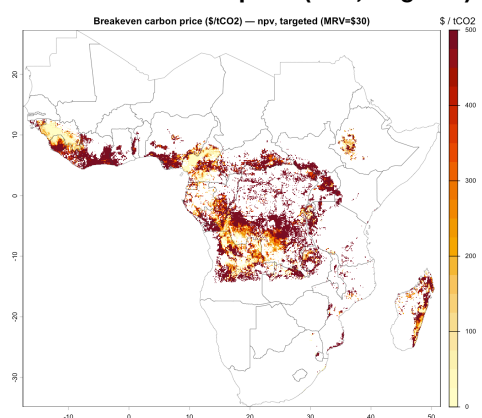
Crop-weighted GM (\$/ha, NPV, targeted)



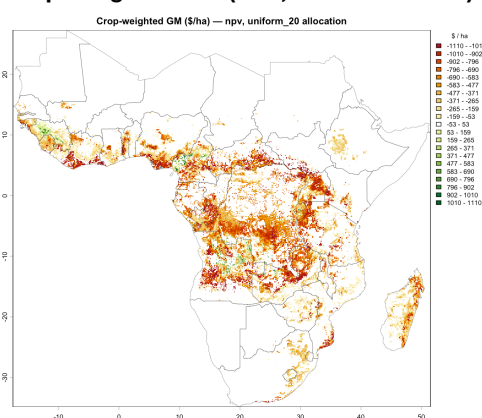
Profitable mask (NPV, targeted)



Breakeven carbon price (NPV, targeted)



Crop-weighted GM (NPV, uniform 20 t/ha)



7.2 SSA-wide totals under different assumptions (refreshed 2026-09-01)

Source: docs/tables/output-ssa-summary.csv — one row per regime × allocation. Refreshed 2026-09-01 after the ESROC-anchored citation review landed (Arrhenius temperature factor, pH peak 6.0, net-export CDR accounting, yield sweep capped 1.8×), on top of the 2026-05-26 defaults (grain 50 μm, MRV \$20/tCO₂, year-1 agronomic summed over 5-yr horizon, CGIAR-CSI Global Aridity Index, per-crop EcoCrop max-pH).

Regime	Allocation	Total GM (M\$)	Agro (M\$)	Basalt (M\$)	CDR rev. (M\$)	CDR (Mt)	Profitable (Mha)	CDR on profitable (Mt)
year-1	targeted	+12,371	30,220	21,545	4,483	34.5	7.4	25.6
NPV	targeted	−5,926	12,030	21,545	3,560	34.5	2.6	16.9
equilibrium	targeted	+1,772	6,044	7,364	3,248	25.0	6.5	12.4
year-1	uniform 10	+14,357	30,220	75,918	11,512	88.6	10.6	1.1
NPV	uniform 10	−10,548	5,495	75,918	9,143	88.6	3.2	0.4
equilibrium	uniform 10	−3,774	6,044	75,918	17,967	138.2	7.8	15.7
year-1	uniform 20	+1,422	30,220	151,836	27,978	215.2	8.1	7.6
NPV	uniform 20	−24,285	5,495	151,836	22,218	215.2	1.9	2.4
equilibrium	uniform 20	−11,850	6,044	151,836	40,188	309.1	6.3	29.2
year-1	uniform 50	−25,337	30,220	379,591	92,169	709.0	6.7	50.5
NPV	uniform 50	−55,929	5,495	379,591	73,196	709.0	2.0	19.3
equilibrium	uniform 50	−36,049	6,044	379,591	107,153	824.2	4.9	63.2

Regime	Allocation	Total GM (M\$)	Agro (M\$)	Basalt (M\$)	CDR rev. (M\$)	CDR (Mt)	Profitable (Mha)	CDR on profitable (Mt)
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Read across: all rows are now on the **net-export** CDR basis (the first-application acidity haircut removes ~70 % of gross first-round CDR; the equilibrium haircut is much smaller), which is why the CDR columns and profitable areas are far leaner than the pre-2026-09-01 gross-basis table. The **year-1 targeted** row leads among the targeted runs at **+\$12.4 B** total SSA-wide GM — the horizon-summed agronomic side scales with 5× annual agronomic returns. **Year-1 uniform 10** is the highest absolute total (\$14.4 B) because basalt is applied across all cropland regardless of acidity and the per-crop max-pH cliffs make most of that cropland agronomically responsive. The **NPV-targeted** row is negative (–\$5.9 B): the 10 % discount rate plus the net-export haircut eat both revenue streams. **Equilibrium-targeted** stays positive (+\$1.8 B) at a 25.0 Mt CDR footprint. Uniform-50 stays negative across all regimes — 50 t/ha on every cropland pixel is excess basalt where it isn’t needed, and the deep-tail cost dominates. The full per-country and per-crop breakdowns are in docs/erw-results.pdf. The same CSVs feed erw-supply.R’s ranking.

8. Parameters and constants

See docs/erw-parameters.md for the full catalogue with citations, source URLs and DOIs. The blocks in §6 above repeat only the constants that materially set the headline result; the parameters doc is authoritative for everything else.

Quick reference of the six **decision variables** (the only knobs swept by erw-8):

Decision variable	Default	Where set	Sweep range
CaO_frac, MgO_frac (feedstock)	0.10 / 0.07	erw-3	not currently swept
grain_size_um	50 µm	erw-3	10 / 30 / 50 / 100 / 200 / 500 µm
allocation	LiTAS-targeted	erw-7, erw-8	+ uniform 10 / 20 / 50
CARBON_PRICE_USD_T	\$150 / tCO ₂	erw-7, erw-8	swept jointly with crop + basalt price
MRV_COST_USD_T_CO2	\$20 / tCO₂	erw-7, erw-8	0 – 80 \$ / tCO ₂
DISCOUNT_RATE_PCT	10 %	erw-7	NPV regime only

9. Validation and known issues

What has been sanity-checked

- B1 pH map matches the qualitative expectation for SSA (acidic in humid tropics, alkaline in Sahel and parts of southern Africa).
- B3 CDR-yield map is strongest in the humid tropics (Guinea, Congo basin) and suppressed in the Sahel — the expected pattern given climate scaling and the aridity-driven pedogenic-C deduction.

- B4 delivered-price map shows the expected geography — cheap near basalt outcrops (Ethiopian highlands, East African Rift, Cameroon volcanic line, Madagascar, southern Africa), capped at \$50 / t across West Africa and the Sahel.
- B4 source-mask polygon counts match the GLiM v1 catalogue (5,266 vb + 1,189 pb in SSA).

Open issues

Issue	Severity	Block	Mitigation
Bayesian yield arm not fit	high (limits scenario realism)	B6	run <code>erw-yield-fit.R</code> once <code>brms</code> + <code>Stan</code> are installed; the trial dataset is the bottleneck, not the code
Pedogenic $\square$ acidity deduction overlap	low (conservative)	B3	1.28 % of pixels are both arid and acid; the sequential ordering over-deducts F·S·p there — documented, not restructured
Industrial by-products not modelled	medium (regional realism)	B4	requires a separate inventory of steel mills, cement plants, mine tailings
Ultramafic outcrops excluded from feedstock	low (intentional)	B4	mark them explicitly in a future “rejected feedstock” overlay
MRV cost is constant \$20 / tCO ₂ regardless of scale or protocol	low	B7	sweep already covers 0 – 80 \$; consider scale-dependence in a future revision
Pedogenic-C deduction uses MAP only, not full AI	low	B3	switch to AI = MAP / PET when a PET surface is available
14-row trial dataset is prior-dominated	medium	B6 (Bayesian arm)	append trials as new evidence appears (Beerling group, InPlanet, Flux trials)
Single-currency model	low	all	USD throughout; not a problem for ex-ante analysis

10. How to reproduce

10.1 Environment

R 4.5+, with the `geodata`, `terra`, `sf`, `here`, `brms`, `malariaAtlas` packages
 Rscript erw/erw-0-install-packages.R

LaTeX is needed only for the framework + pipeline figures (`pdflatex` + `pdftocairo`).

10.2 Run order

1. inputs and working grid
 Rscript erw/erw-1-layers-soilgrids.R
 Rscript erw/erw-2-layers-spam.R

2. basalt requirements

```
Rscript erw/erw-3-basalt-requirements.R
```

3. crop response

```
Rscript erw/erw-4-crop-parameters.R
```

```
Rscript erw/erw-5-crop-yield-loss-function.R
```

4. CDR + cost surfaces

```
Rscript erw/erw-9-cdr-yield.R
```

```
Rscript erw/erw-energy-country.R
```

```
Rscript erw/erw-fetch-friction-surface.R # one-off – pulls MAP friction
```

```
Rscript erw/erw-basalt-access.R
```

5. (optional) Bayesian yield arm

```
Rscript erw/erw-yield-fit.R
```

```
Rscript erw/erw-yield-predict.R
```

6. profitability, supply, sensitivity

```
Rscript erw/erw-7-profitability-function.R
```

```
Rscript erw/erw-supply.R
```

```
Rscript erw/erw-8-sensitivity-analysis.R
```

7. render maps, results, and rebuild figures

```
Rscript erw/erw-plot-maps.R
```

12 indicator maps

```
Rscript erw/erw-10-profitability-maps-tables.R # profitability maps + 3 CSV tables
```

```
pdflatex docs/erw-pipeline-flow.tex \
```

```
&& pdftocairo -svg docs/erw-pipeline-flow.pdf docs/erw-pipeline-flow.svg
```

```
pdflatex docs/erw-conceptual-framework.tex \
```

```
&& pdftocairo -svg docs/erw-conceptual-framework.pdf docs/erw-conceptual-framework.svg
```

8. rebuild documentation PDFs

```
bash docs/build-erw-model.sh
```

```
bash docs/build-erw-results.sh
```

```
bash docs/build-erw-tutorial.sh
```

10.3 Expected runtimes (Apple Silicon, single core)

Stage	Time
erw-1, erw-2 (downloads + resampling)	~30 – 60 min first time, cached afterwards
erw-3, erw-9	a few minutes each
erw-fetch-friction-surface	5 – 10 min (one-off download)
erw-basalt-access (costDist)	~10 min
erw-7, erw-8	minutes – hours depending on sweep depth
erw-plot-maps (12 maps)	~15 min
erw-10-profitability-maps-tables (48 maps + 3 CSVs)	~5 min

11. Gaps and roadmap

Near-term (within current month)

1. Fit the Bayesian yield arm (`erw-yield-fit.R`) once `brms` + `Stan` are installed locally.
2. ~~Run `erw-7` end-to-end~~ — done (2026-05-19 first run; refreshed 2026-09-01): 276-raster profitability stack `{crop}_{regime}.tif` under `economics_erw/{allocation}/`.
3. ~~Run `erw-supply` and `erw-8`~~ — done (both re-run 2026-09-01 at the current defaults).
4. Add indicator maps for the supply curves and the year-1 / NPV / equilibrium profitability comparison.

Medium-term

5. **Industrial by-product feedstock layer.** Inventory of steel mills (GEM Steel Plant Tracker), cement-clinker plants (Global Cement Directory), and active mine tailings (USGS / national geological surveys). Add a point-source feedstock layer and feedstock- chemistry table; rerun B4 with the union of mafic outcrops and by-product points.
6. **Cross-country MRV cost differentiation.** Some jurisdictions (Brazil, Kenya) have simpler smallholder-aggregator pathways; the flat \$30 / tCO₂ assumption over-estimates cost for those.
7. **Aridity index using AI = MAP / PET** instead of MAP-only for the pedogenic-C deduction. Requires a PET surface (CGIAR-CSI or computed from WorldClim).

Long-term

8. **Multi-feedstock decision rule.** Currently feedstock chemistry is fixed; the user chooses one. Generalise to a per-pixel cheapest-eligible-feedstock rule given a chemistry inventory.
9. **Time-resolved deployment scenarios.** Year-by-year deployment under supply caps, carbon-price paths, and yield-learning curves.
10. **Smallholder-aggregator economic model.** Costs and revenues at the aggregator level (transport, MRV pooling, finance) rather than the per-pixel “farmer” level.

12. References

The full bibliography is in `docs/erw-review.md`. The most load-bearing citations are:

- Beerling, D. J. et al. (2020). *Potential for large-scale CO₂ removal via enhanced rock weathering with croplands*. **Nature** 583, 242 – 248.
- Strefler, J. et al. (2018). *Potential and costs of carbon dioxide removal by enhanced weathering of rocks*. **ERL** 13, 034010.
- Lewis, A. L. et al. (2021). *Effects of mineralogy, chemistry and physical properties of basalts on carbon capture potential and plant-nutrient element release via enhanced weathering*. **Applied Geochemistry** 132, 105023. doi:10.1016/j.apgeochem.2021.105023. (Reference for the reactive-fraction calibration used in `erw-3`.)
- Hartmann, J. & Moosdorf, N. (2012). *The new global lithological map database GLiM*. PANGAEA doi:10.1594/PANGAEA.788537.
- Weiss, D. J. et al. (2020). *Global maps of travel time to healthcare facilities*. **Nature Medicine** 26, 1835–1838. doi:10.1038/s41591-020-1059-1. (Source of the Malaria Atlas Project 2019 motorised friction surface used in `erw-basalt-access`; the citation-review of 2026-09-01 corrected an earlier attribution to Weiss et al. 2018.)

- White, A. F. & Blum, A. E. (1995). *Effects of climate on chemical weathering in watersheds*. **Geochimica et Cosmochimica Acta** 59 (9), 1729–1747. (Source of the $E_a = 68.8$ kJ/mol silicate activation energy in the Arrhenius temperature factor of $erw-9$.)
- Bertagni, M. B. & Porporato, A. (2022). *The carbon-capture efficiency of natural water alkalization*. **Science of the Total Environment**. (Capture-efficiency collapse below pH 5 — one anchor of the pH-6.0 peak in $erw-9$; with Holden et al. 2024, *Science of the Total Environment*, and Power et al. 2025, *Frontiers in Climate*.)
- Mercer, L. et al. (2024). *Measurement, reporting and verification costs for enhanced weathering*. LSE Grantham Research Institute. (EW MRV \$15–71/tCO₂ — anchor for the \$20/tCO₂ MRV default in $erw-7$.)
- Aramburu Merlos, F., Silva, J. V., Baudron, F. & Hijmans, R. J. (2023). *Estimating lime requirements for tropical soils: Model comparison and development*. **Geoderma** 432, 116421. (Reference for the LiTAS lime-equivalence recipe used in $erw-3$.)
- Kanzaki, Y., Planavsky, N. J. & Reinhard, C. T. (2023). *New estimates of the storage permanence and ocean co-benefits of enhanced rock weathering*. **PNAS Nexus** 2 (4), pgad059.
- Beerling, D. J. et al. (2024). *Enhanced weathering in the US Corn Belt delivers carbon removal with agronomic benefits*. **PNAS** 121 (9), e2319436121.
- Haque, F. et al. (2025). *Agronomic Performance of Enhanced Rock Weathering in a Tropical Smallholder System: A Maize Trial in Kenya*. CDRXIV preprint 410 (Flux Carbon / UNCCD).
- Dupla, X., Bertagni, M. B. & Grand, S. (2025). *Three Years of Field Trials Indicate a Sustained Enhanced Rock Weathering Signal with Limited CO₂ Removal*. **Environmental Science & Technology** 59 (48), 25751–25764.

This document is the canonical state of the model. When you change a script, a constant, a data source, or an output, edit the relevant block under §6 and add a dated entry to §1. Anything that is not in this document — even if it lives in the code — is not part of the model's documented behaviour.